

Danny Rollo

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Available now

EDUCATION

Northeastern University M.S. in Computer Science B.S. in Computer Science & Mathematics	Boston, MA Sept 2024 – April 2026 Sept 2021 – May 2025
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PUBLICATIONS

Rollo, D. “Lightweight CNN Encoders for Nucleus Segmentation.” *MICCAI 2026*. **Accepted — poster.**

TECHNICAL SKILLS

Languages & Core Libraries: Python, Go, TypeScript/JavaScript, SQL, pandas, NumPy
Deep Learning: PyTorch, timm, CNNs, U-Net, Vision Transformers (Swin), Transformers & attention
Machine Learning: scikit-learn, InterpretML (EBM), gradient boosting, PCA, SVM, collaborative filtering
LLM & Agentic Systems: Claude and OpenAI APIs, MCP servers, multi-agent orchestration, prompt engineering, RAG, SentenceTransformers
Infrastructure: Docker, AWS (S3), FastAPI, MongoDB Atlas, Git

SELECTED PROJECTS

- Lightweight CNN Encoders for Nucleus Segmentation** PyTorch, timm, PanNuke, MoNuSeg
- Designed and ran a controlled encoder ablation isolating encoder choice in U-Net segmentation, holding decoder, augmentation, and optimization schedule fixed across VGG, ResNet, and Swin-T backbones.
 - Established that a 34M-parameter VGG U-Net (Dice 0.851 on PanNuke; 0.796 on MoNuSeg) matches a purpose-built medical transformer and is Pareto-dominant on inference cost.
- Two-Tower Recommendation with Cold Start Ablation** PyTorch, BPR loss, Yelp Open Dataset
- Found context features alone (day-of-week, distance, restaurant temporal profiles) outperform the full model on cold-start users by 4.2 NDCG@10 points, while content features remain essential for cold restaurants — an asymmetry in feature importance across cold-start regimes.
 - Designed a leave-one-out onboarding evaluation that prevents information leakage when a cold user’s only signal comes from held-out reviews.
- RailMax: Amtrak Service Optimizer** InterpretML, GTFS, Census ACS, Folium
- Modeled the US Amtrak network as a directed weighted graph (edges = weekly trips between station pairs) and engineered node-level features including destination-population-weighted reachability and eigenvector centrality.
 - Trained an Explainable Boosting Machine ($R^2 = 0.76$, RMSE 0.84 log scale) under 5-fold stratified group cross-validation with k-means geographic clustering to prevent spatial leakage.

EXPERIENCE

- Teradata — AI/ML Product Management Analytics Intern** June 2025 – May 2026
- Authored the business case and led the design of a context-aware AI Notebook Assistant for VantageCloud AI Studio, enabling intelligent code generation, contextual suggestions, and integration with company libraries and documentation.
 - Built frontend prototypes and interaction specs for agentic notebook assistant features, translating LLM capability constraints into user-facing product decisions.
- PlateMate — Co-Founder & Chief Technology Officer** May 2025 – Present
- Lead ML architecture and engineering for a food recommendation platform: hybrid item-based collaborative filtering plus semantic vector search over restaurant reviews.
 - Built a RAG retrieval pipeline with 768-dimensional `a11-mpnet-base-v2` embeddings on MongoDB Atlas Vector Search, parsing natural-language dish queries into semantic and structured geo-aware filters.
- LineVision Inc. — Software Engineer Co-op** Jan 2024 – June 2024
- Full-stack owner of an internal web application; defined features with stakeholders, expanded APIs and database schema, maintained the application end to end.
 - Re-architected a data pipeline for staff scientists, achieving a 1000× speed improvement.
- Generate Product Development — Project Lead, Technical Lead, Software Engineer** Jan 2023 – May 2025
- Led 14 engineers and designers to deliver a production mobile application; defined requirements with external stakeholders.
- Khoury College of Computer Sciences — Lead Teaching Assistant** 2022 – 2025
- Fundamentals of Computer Science I & II · Led weekly two-hour laboratory sections of 40+ students across six semesters, held 1-on-1 office hours, and graded assignments.

HONORS & AWARDS

Rookie of the Year, HackBeanpot · Students of the Year Runner-Up, LLS · Eagle Scout, Scouts BSA